

RESPONSE TO THE R40 REFEREE REPORT

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This response identifies the mathematical extensions, new executed certificates, corrected evidence classifications, and remaining numerical questions relative to the September 25, 2026 report. It accompanies the complete R42 article, technical supplement, computation report, and reproducible proof objects.

1. OVERVIEW

We thank the referee for distinguishing a complete numerical guarantee from a practically useful certificate, and a deterministic separation from a randomized optimality interval. The revision addresses those distinctions by adding results and executions, not by relabeling old evidence or changing the common-policy objective. In particular, it adds an exact reduction for an atomic continuum class with exogenous dynamics, independently checked long-horizon multi-action closures, a fully instantiated density-transfer budget, finite-horizon repair at discount factor one, and an exact zero-tolerance characterization. The prior nonlinear finest-object arithmetic audit is incorporated with its actual provenance. All earlier sources and results are retained.

The new exact-reduction class is not claimed to include the original controlled affine and bilinear maintenance models. Their positive randomized gaps remain displayed. The new service economies have designed, not estimated, primitives. The response below treats these as scientific distinctions rather than concealing them behind an overall success label.

2. POINT-BY-POINT RESPONSE

F1: Theory does not cover the executed atomic models

The new section on exact atomic aggregation gives two explicit identities: pointwise probabilities of successor cells and a measure-valued pushforward identity for the cell reference laws. Under exogenous transitions and cellwise primitives, these imply equality of the original unrestricted measurable common-policy optimum and a finite-state optimum. The proof averages arbitrary competing policies and lifts a finite optimum pointwise; it does not presume density of a policy sieve. The executed continuum economy has two continuously valued coordinates evolving by nonlinear, measure-preserving shears and a finite atomic shock in the regime. Its transition law is never smeared. The separate density experiment executes the original reset-transfer theorem with nonzero operator errors and both adjusted finite problems. Thus the revision contains both an executed atomic theorem and an executed reset theorem. The original action-dependent models remain outside those sufficient structural identities; we do not imply otherwise.

F2: Exponential completeness is not practical closure

The regret LP derived from the aggregation theorem has $(m + 1)nT$ variables and $2nT$ equalities, with sparse size $O(nT(m + s))$. It preserves all restart constraints and removes policy branching in this class. All twelve registered nonzero-cost cases meet an absolute certificate target of 10^{-4} , with actual gaps below 9×10^{-8} ; the largest has 64 states, 64 dates, and three actions. The general lattice and branch-and-bound results remain intact, including

their exponential worst-case burden. The new undiscounted repair explicitly states its weaker $O(M^{-1/T})$ fixed-horizon rate. We do not present that rate as a practical solution to the generic combinatorial problem.

F3: The McCormick comparator dominates on hard prior rows

The full original paired comparison remains in the main article. We do not claim that interval Bellman search outperforms it. The structural experiment instead compares the eliminated regret LP with an unreduced Bellman LP under ordinary HiGHS presolve and the same information, tolerance settings, and limit. Both represent the identical exogenous problem. Some small cases favor the unreduced LP; larger rows often favor elimination. All timings are reported, with exact certification and independent checking separated. The improvement is a problem reduction and verified LP closure, not an asserted invention of a superior general-purpose optimizer. A direct comparison against the earlier endogenous-transition cases would not be same-object and is not made.

F4: All positive primary randomized intervals remain open

The abstract and main tables now say explicitly that all 30 positive-cost primary intervals remain open. The article adds median and maximum relative widths at each horizon and signed guaranteed savings $L_D - U_R$. The supplement lists all 42 signed savings, including nonpositive cases. The original 27 strict comparisons are preserved with the correct inequality direction. Tight randomized optimality is established on the new structural economies, not retroactively attributed to the primary cohort. Further narrowing of the original 30 intervals is not reported as completed.

F5: Prices and spatial refinement do not prove atomic convergence

The new atomic theorem neither assumes an exhaustive finite price portfolio nor infers convergence from mesh refinement. It proves an exact aggregation equivalence under stated kernel identities. The older local and support lower bounds remain valid one-sided certificates in their original models and are not promoted to a convergent solver for those models. We retain the original target rather than optimizing an uncontrolled relaxation.

F6: Nonlinear widths, loose allowance, and incomplete finest verification

The successful R41 independent audit of the complete $T = 32, N = 256, \varepsilon = 1/2$ object is now reflected in the main table and the theorem-to-checker map. Its actual source, proof and verifier hashes, run identifier, inequality count, and result are retained. The same interval remains 25.13% wide; this audit is not an optimality improvement. The four tighter $\varepsilon = 0.05$ nonlinear rows remain lower-only and are never paired with a policy certified at $1/2$. The new nonlinear-shear continuum economy has exact paired certificates at 0.01 and 0.05, but its exogenous structure is identified explicitly. General convergence and tight closure of the original action-dependent box representation are not claimed.

F7: Too few states, actions, and transition structures

The new predeclared suite includes 8, 16, 32, and 64 states; horizons 8, 16, 32, and 64; three or five actions; banded, cyclic, and dense transitions; discount factors 0.95, 0.99, and 1; and

allowances 0.01 and 0.05. All twelve cases are retained. The registration preceded execution but did not freeze the final source code, so it is described as a predeclared structural experiment, not the earlier fully code-frozen holdout or empirical validation. Its breadth is within the proved exogenous class, not across unrestricted economic kernels.

F8: Witness-driven theory is disconnected from a difficult operating solve

The original witness theorem and its full proof are retained. The new allowance ladder supplies a date-specific witness residual condition that can also be checked at discount factor one. In the structural class, regret cancellation eliminates the need to approximate the operating value before solving the revision problem: the LP uses immediate reward losses and the transition matrix directly. This is a structural removal of the oracle burden, not an example in which an exact operating solve is intrinsically difficult. The previous witness test table remains a test of the theorem and is not relabeled a hard operating-approximation experiment. A genuinely difficult approximate operating problem with useful end-to-end constants remains a separate empirical test not supplied by these service cases.

F9: Transfer constants and initial-reference measures are restrictive

The density experiment constructs the reference measures and initial mixture exactly and computes operator error $\alpha = \zeta/2$, probability total variation $\zeta/4$, zero stage/terminal approximation errors, every propagated operating and cost error, and both adjusted finite certificates. The tables report η_+ , η_- , $2b_h$, repair cost, total sufficient width, and measured width separately. The largest perturbation fails the tightening condition and is retained. The exact aggregation reference remains valid at that same parameter; this exposes the weakness of a sufficient error bound rather than hiding it. No dimension-free generic spatial approximation rate is asserted.

F10: Economic primitives remain uncalibrated

The new service economy is an explicit model of revising installed intensities under an exogenous demand process, with stated implementation units and all-restart operating protection. It is not estimated from data, and no monetary welfare, empirical prevalence, or confidence-set coverage is invented. The response to the referee's alternative request for a stronger methodological contribution is the exact atomic reduction, its executed larger closures, and the extension of the repair theory. State- and action-dependent administration charges are now accommodated directly in the structural LP when they leave operating dynamics unchanged. Charges that affect rewards or transitions require recertification of those primitives. A calibrated institutional application remains additional evidence, not something inferred from synthetic parameters.

F11: Independent coverage is incomplete on the hardest objects

The complete finest nonlinear object is now marked with its successful independent R41 audit, not merely a structural or hash check. Every new structural and adjusted-density certificate receives a fresh optimizer-free rational reconstruction of its primitives, all action probabilities, all restart regrets, implementation cost, and dual residual bound. The supplement distinguishes mathematical reductions, independent arithmetic, constructor checks, protocol registration, and publication integrity. It does not equate the source-level pushforward proof with a machine-checked theorem or a hash with arithmetic validity.

F12: Novelty relative to constrained-MDP and global methods

The literature discussion now includes an explicit compatible occupation-flow formulation. Shared policy products $y_{tia}^s = z_{ti}^s p_{tia}$ show precisely where independent restart measures would enlarge the policy class. Under exogeneity the state occupancies are fixed; the regret LP eliminates those redundant flows while preserving the common policy. The atomic theorem proves equality for unrestricted Borel competitors using both kernel identities, rather than assuming a quantized policy class. Existing finite LP approximation results by Dufour and Prieto-Rumeau and by Saldi are credited. We claim neither McCormick envelopes, simplex, aggregation in general, nor verified residual arithmetic as new inventions.

F13: Local-LP work may not justify its contribution

The new structural closures do not use the local-support LP component, so they do not attribute their tightness to extra local prices or cell work. The earlier component and its full work records remain in the retained article and supplement. Its actual connected lower-bound improvement is still distinguished from global closure. No new equal-budget local-versus-global ablation has been executed, and the revision does not invent marginal runtime comparisons from nonmatched historical runs. This specific empirical comparison remains unperformed; the component is preserved as a one-sided accelerator rather than a necessary part of the exact structural solver.

F14: Scope excludes undiscounting, zero tolerance, and long horizons

The finite-horizon slack-ladder theorem permits $\beta = 1$ with a constructive Borel repair and explicit modulus, tested on action-dependent kernels. A separate exact proposition solves $\varepsilon = 0$ by cost minimization over operating-optimal actions. The supplement also proves a discounted infinite-horizon tail enclosure with an explicit exact-terminal-value trust boundary and a time-dependent policy class. No stationary-optimality or average-cost conclusion is inferred from it. The old stopped-control derivation and its unmet 0.01 target remain archival, outside the new completed examples. These are extensions of the theory, not a narrower replacement objective.

3. ADDITIONAL PRESENTATION REQUESTS

The abstract identifies both the new structural scope and the original open intervals. Main tables now distinguish zero-cost exactness, randomized optimality widths, and guaranteed deterministic savings. The new work table reports nT , variables, nonzeros, constructor and checker time, proof bytes, and raw memory metadata. The old 60-second branch-and-bound cap remains described as a check between proof-producing iterations. Source registration is not confused with code freezing, the initial law is not treated as empirical, and the density norm convention is explicit in both the construction and table.

Some requested diagnostics cannot be honestly derived from the retained objects alone: a new same-budget local-work ablation, an empirical calibration, and new feasible tight-tolerance policies for the original nonlinear model are not presented as completed. The original witnesses, local-LP records, source hashes, failures, and all original manuscript material remain available. The current main article contains the new proofs and conclusions directly, so the reader need not reconstruct them from a chain of revisions.

4. REVIEW PACKAGE

The entry points are `ECTA_R42.pdf`, `SUPP_R42.pdf`, `RESPONSE_R42.pdf`, and `COMPUTATION_R42.pdf`. The new revision directory contains the sources, exact compressed proof objects, independent checker, full result files, protocol, coverage map, and publication manifest. The new branch descends from the R41 audit and its R40 referee-report parent; earlier branches and historical documents are not overwritten. The next review can therefore distinguish the added scientific claims from unchanged prior evidence.